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PROBLEM & USERS: ECOSYSTEMS ACROSS AN AREA

How could development change area water reserves and watershed conditions across seasons?

REGIONAL GOAL

Seasonal area conditions
Development demands + returns
Municipal reserves + watershed

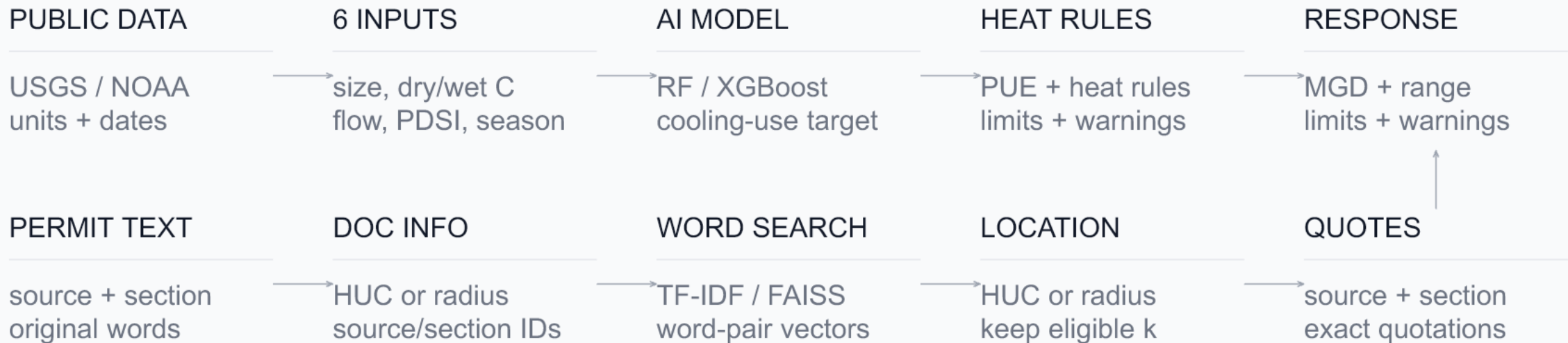
CURRENT PROTOTYPE

Environmental data + cooling model
Physical checks + permit quotes
Regional assessment still planned

Cooling is one case study. Users: municipal planners, researchers, utilities.

SEMANTIC MAP: CURRENT SYSTEM

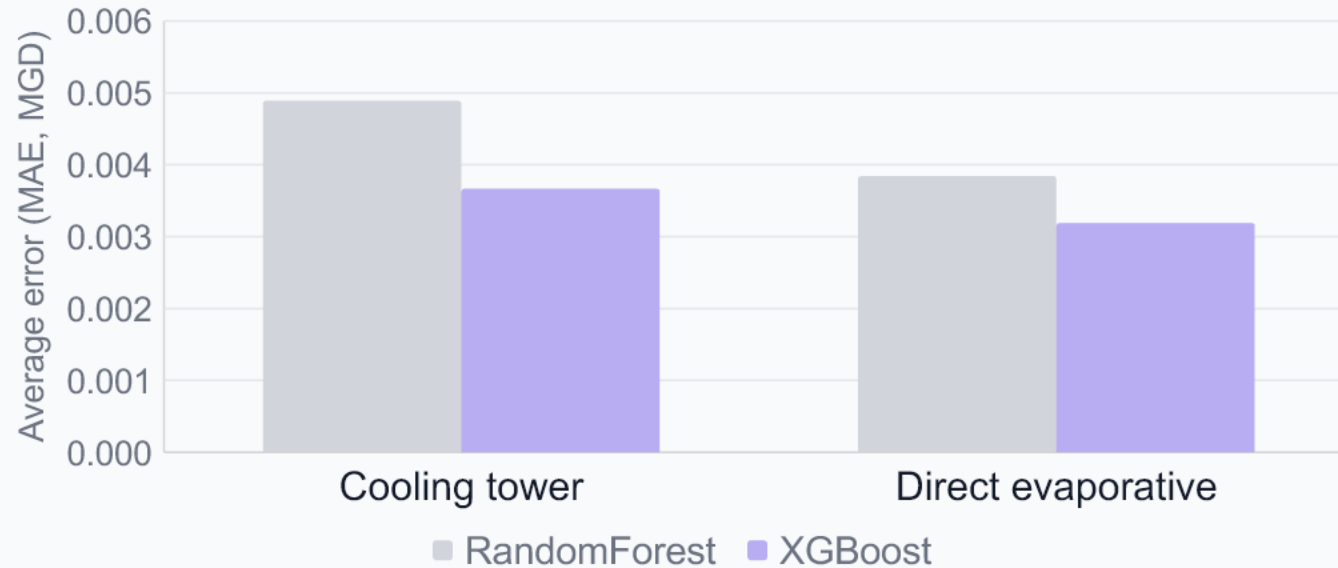
Current API example: IT size, cooling type, watershed (HUC8), month; coordinates optional



Rules check learned numbers. Place metadata filters word vectors. Quotes add context.

E3: 40 MW fault test clips to [0, 0.446484] MGD. E4: four location checks pass.

EXPERIMENT E1: MODEL COMPARISON



KEEP AS A COMPONENT

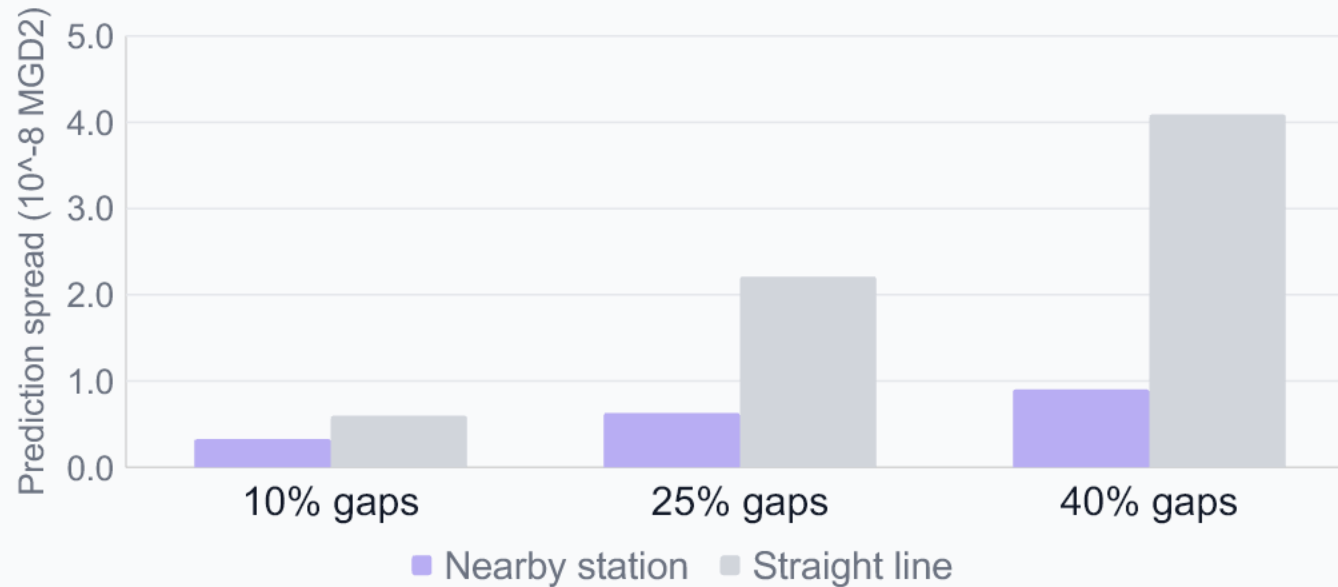
Choose with RMSE on the middle date group. Keep RF for comparison.

Next: measured demand and regional outcome data.

Dates: train 60% / choose + set ranges 20% / test 20%

Dry air cooling: zero water target; R2 is undefined.

EXPERIMENT E2: GAPS & BIASED NEIGHBORS



100%

All main cases passed

Change within +/-0.05 MGD
50 repeats; 121 drought dates

FAILURE: biased neighbors
only 60.3% within limits

Improve: check neighbors and shared outages. The biased test uses a separate simple formula.

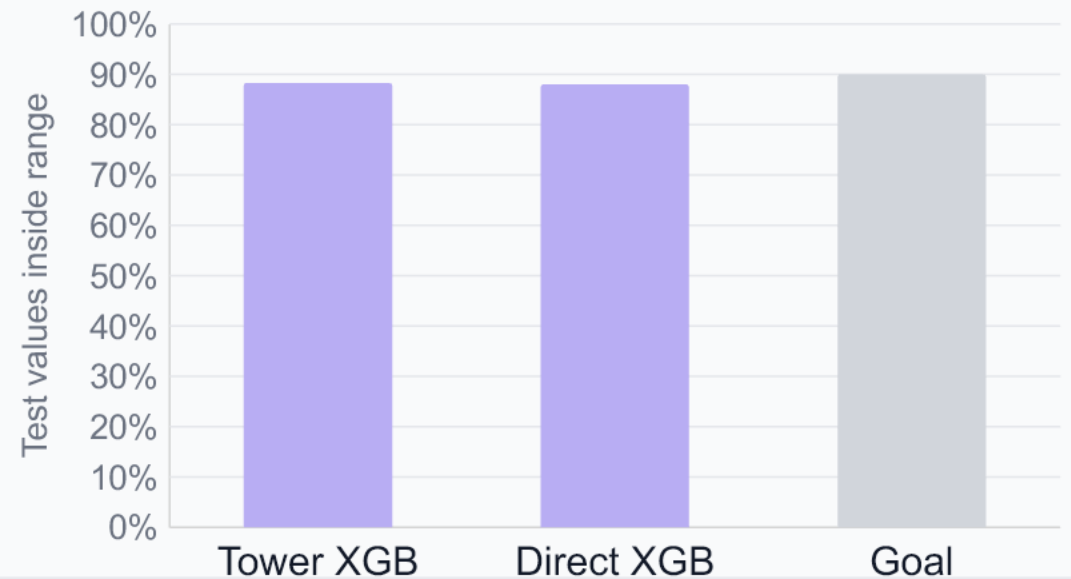
EXPERIMENT E₅: RANGE RELIABILITY

88.3% / 88.0%

RELIABILITY ISSUE: below the 90% goal

Tower R² = 0.998

Good fit does not prove the range is reliable.



Improve: use separate data to choose models and set ranges.

Limited samples; no statistical significance test or independently verified coverage guarantee.

DECISIONS & NEXT QUESTION

KEEP (ADOPT)

E1: lower XGBoost error. E3/E4: safeguards pass.
Keep models, rules, and location checks.

CHANGE (MODIFY)

E2: biased neighbors. E5: ranges miss 90%.
Improve gaps/ranges and add area balances.

REJECT

E5: R2 is fit, not confidence.
Cooling use is not regional ecosystem impact.

DEFER

No neural/LLM benchmark or regional labels yet.

Can we validate seasonal reserve and watershed impacts in an unseen area?

Shaun: code + area integration
Troy: data + grounding
Scott: evaluation + evidence